User Interface Design and Interaction Flows

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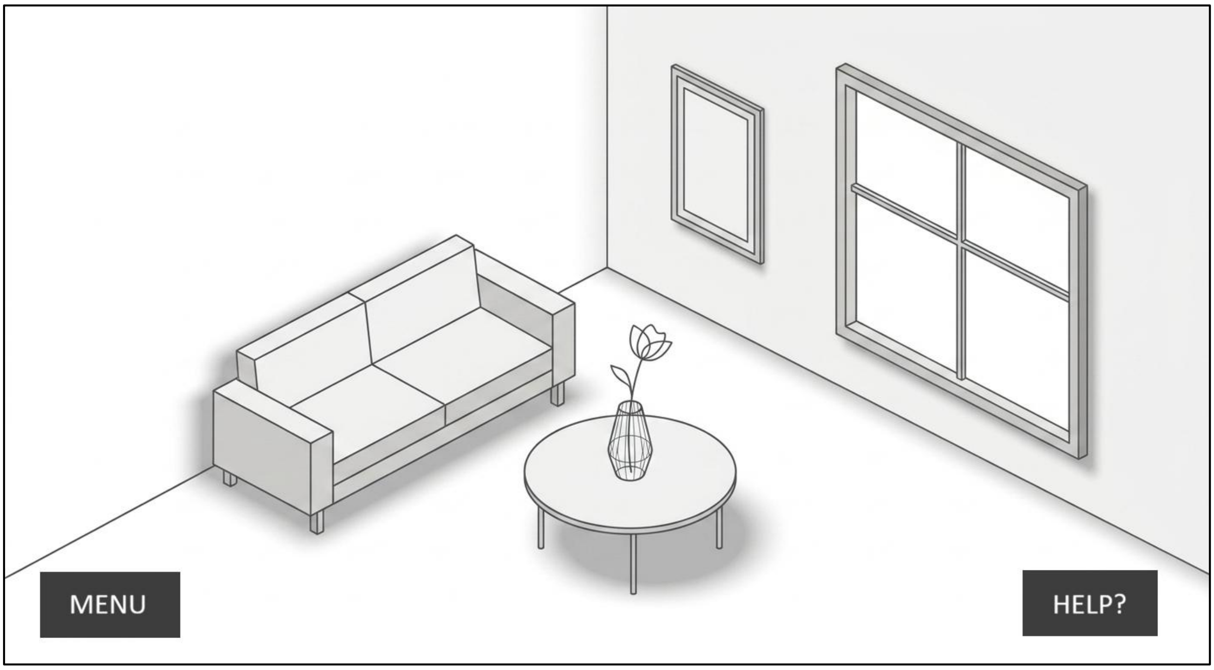
Version : 1.1

1. Object Selection and Texture Change using Text

1.1 Scene 1: Enter the room

Interaction:
Walk / Navigate through the room

Story:
The user enters the virtual interior and moves around the space using walking or navigation controls. They can look around and explore the environment before interacting ith any object. The Menu and Help buttons remain accessible but unobtrusive.

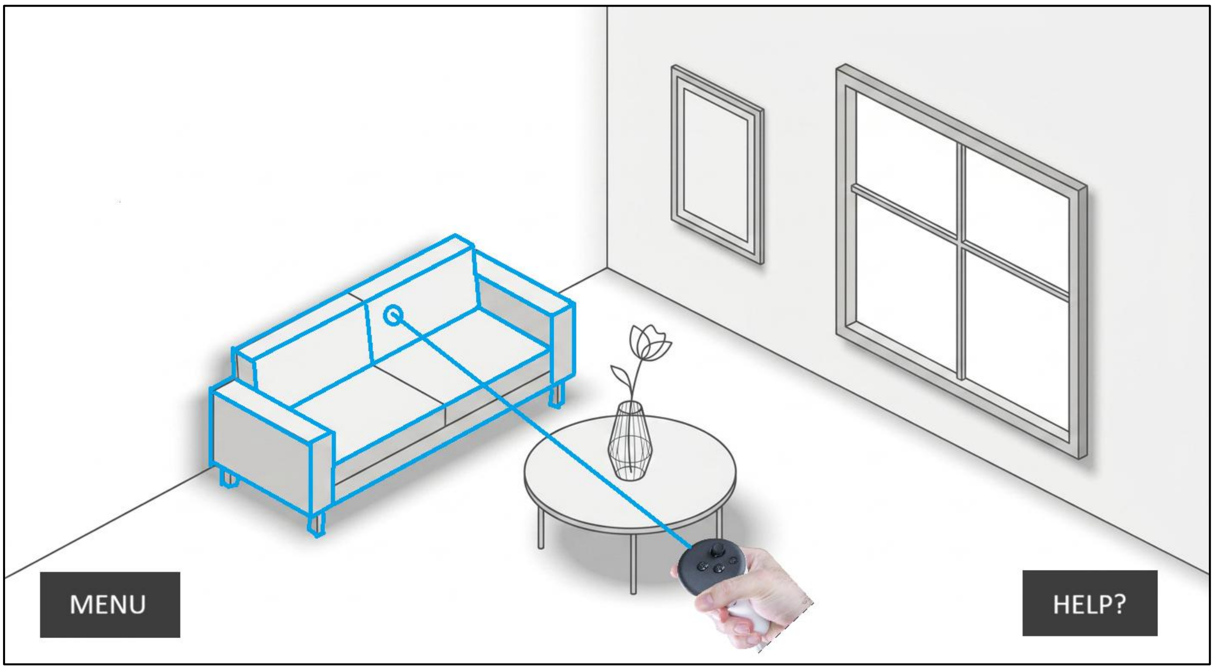


User goal:
Explore the environment and identify objects that can be customised.

1.2 Scene 2: Point at an Object

Interaction:
Point / Hover over an object

Story:
The user points toward an object, such as the sofa. When the user's pointer is over an editable object, the object receives a **subtle light-coloured highlight**. This provides immediate visual feedback that the object is interactive and can be customised.

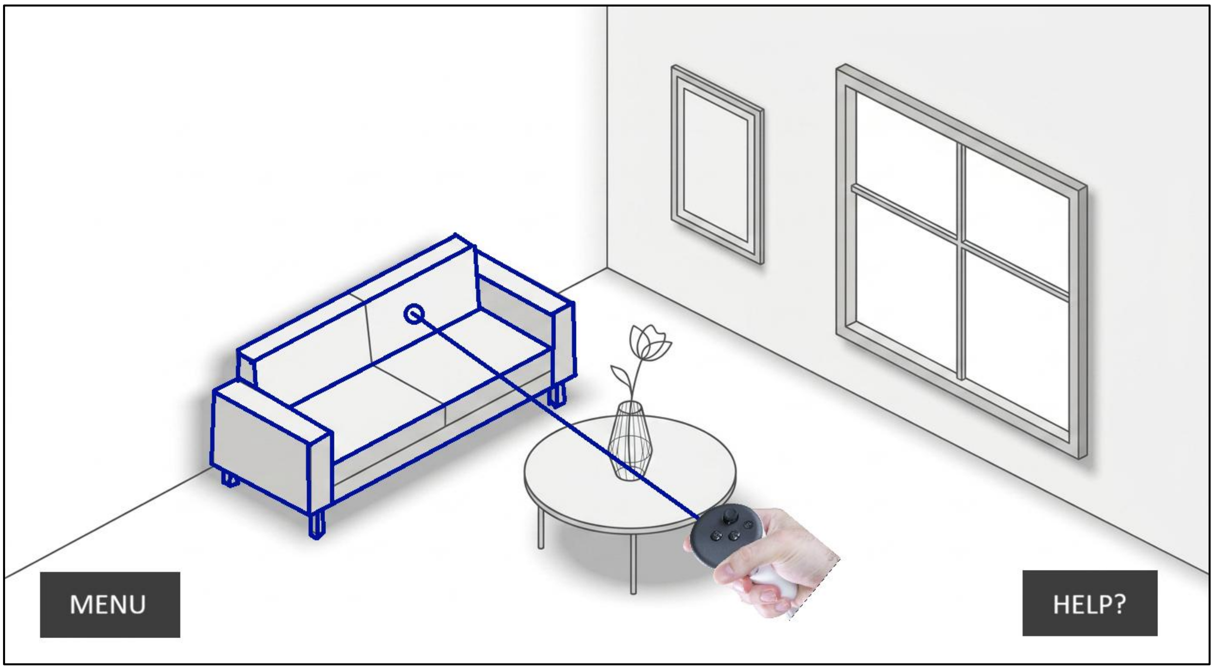


User goal:
Identify which objects in the environment are editable and available for customisation.

1.3 Scene 3: Select the Object

Interaction:
Click / Select the highlighted object

Story:
The user selects the highlighted object. The selected object changes from the **light highlight to a clear dark border**, confirming that the object has been successfully selected. The selection state makes it clear which object the user's next action will affect.

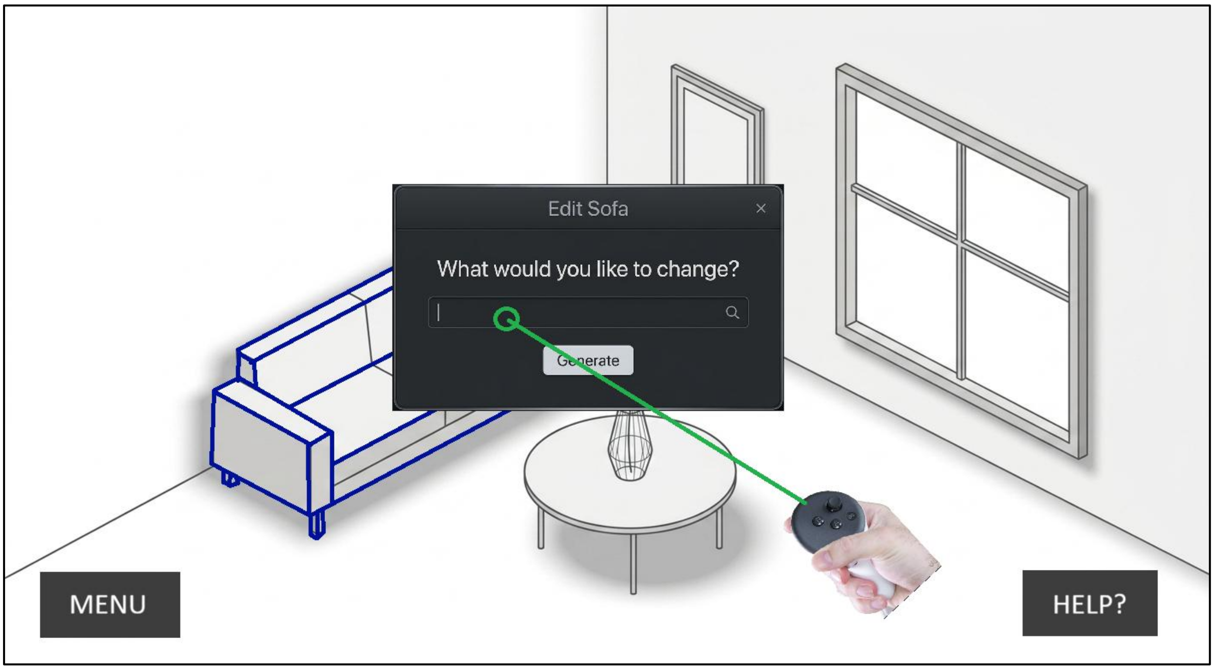


User goal:
Confirm the specific object they want to customise before moving to the editing and AI customisation flow.

1.4 Scene 4 : Customisation Panel

Interaction:
Open / access object customisation

Story:
Once the user selects an object, a **contextual customisation panel** appears near the selected object. The panel clearly identifies the selected object and provides an option to describe the desired change using an AI prompt.



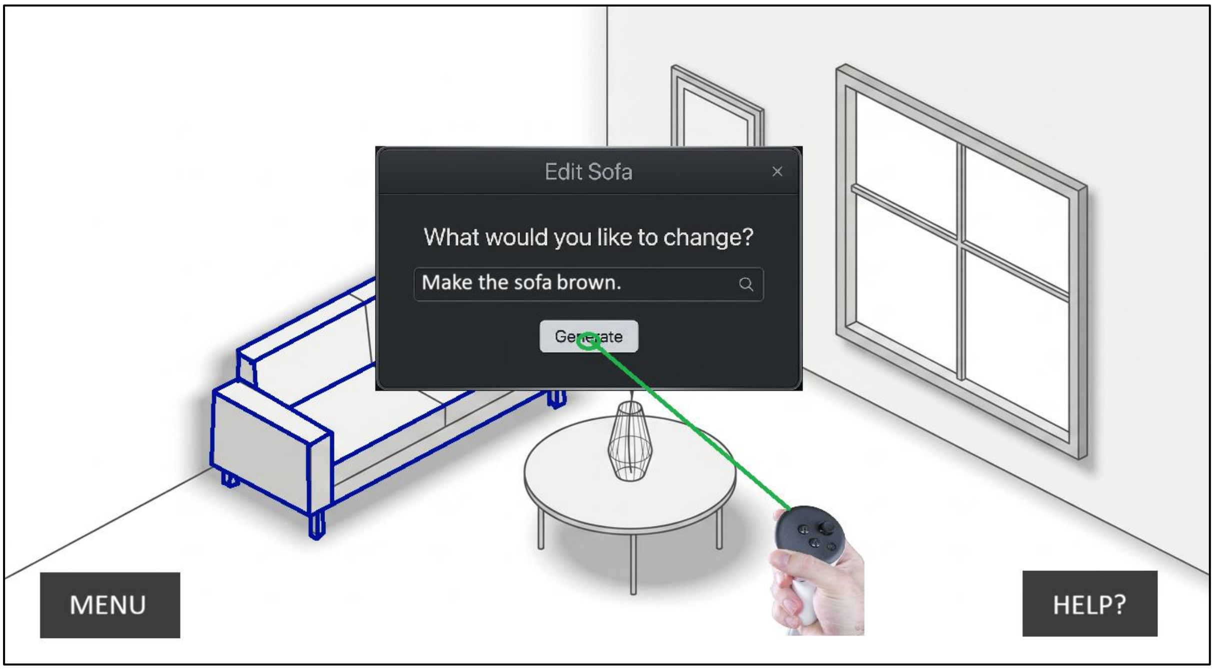
User goal:

Access the editing options and describe what they want the AI to change.

1.5 Scene 5 : AI Processing

Interaction:
Enter prompt → Generate

Story:
The user enters their desired modification and selects **Generate**. The system provides clear visual feedback that the AI is processing the request. The selected object remains identifiable while the system works.



User goal:

Understand that the request has been received and the AI is generating the requested modification.

1.6 Scene 6 : AI Processing

Interaction:
Wait / Monitor AI processing

Story:
After the user clicks **Generate**, the system processes the request. A clear loading or progress indicator tells the user that the AI is working on the requested change.

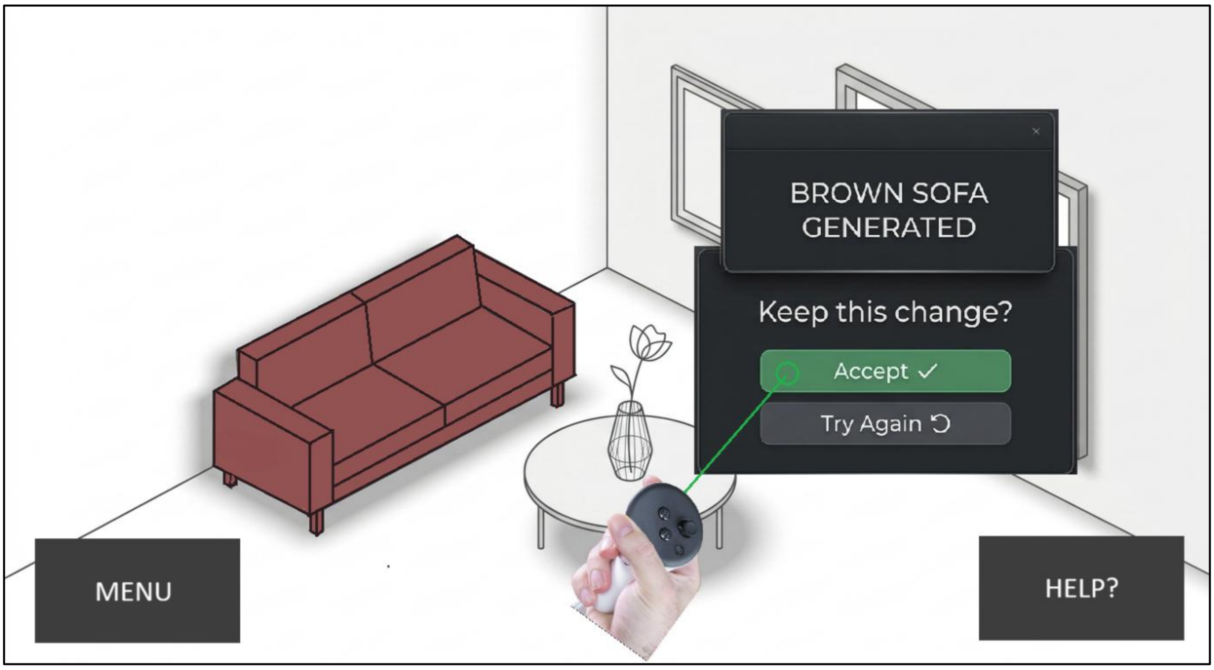
User goal:

Understand that the request has been received and the AI is currently working.

1.7 Scene 7: Generated Result & Review

Interaction:
Review → Accept / Try Again

Story:
The AI finishes processing and displays the updated object in the room. The user can clearly see the generated change. A **small contextual pop-up** appears beside the object without covering or obstructing the generated result. The user can review the change and choose to **Accept** it or **Try Again**.



User goal:

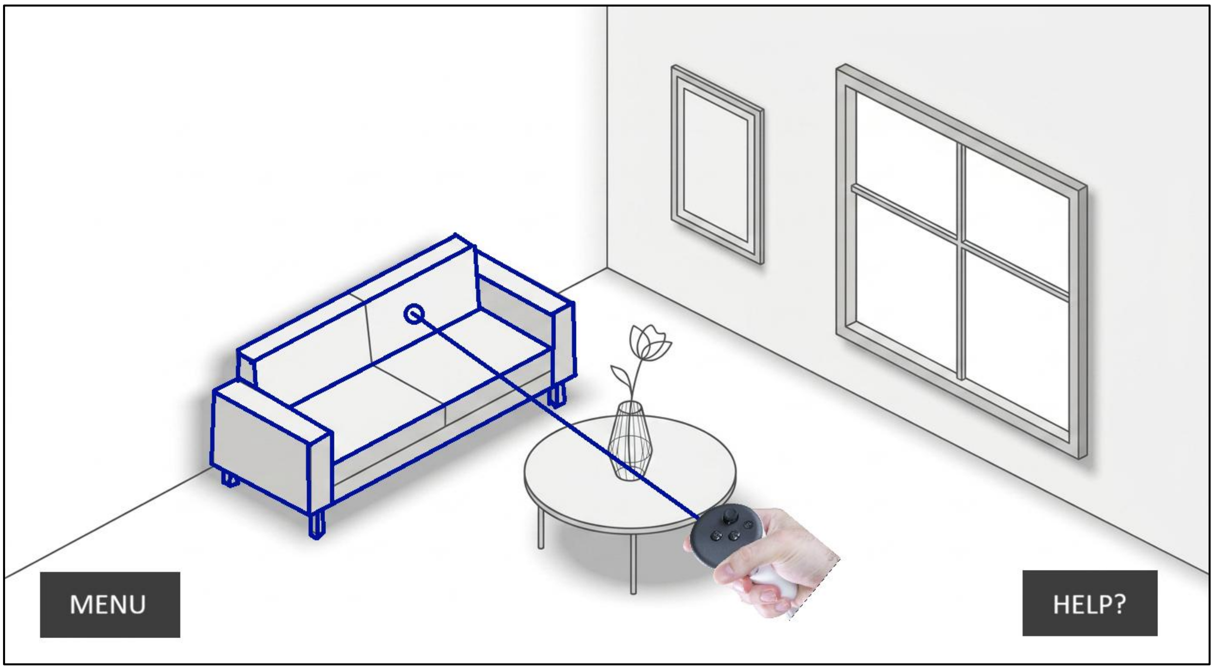
Review the generated change and confirm whether to apply it.

2. Object Selection and Move the Object using the controller

2.1 Scene 1: Select the Object

Interaction:
Click / Select the highlighted object

Story:
The user selects the highlighted object. The selected object changes from the **light highlight to a clear dark border**, confirming that the object has been successfully selected. The selection state makes it clear which object the user's next action will affect.



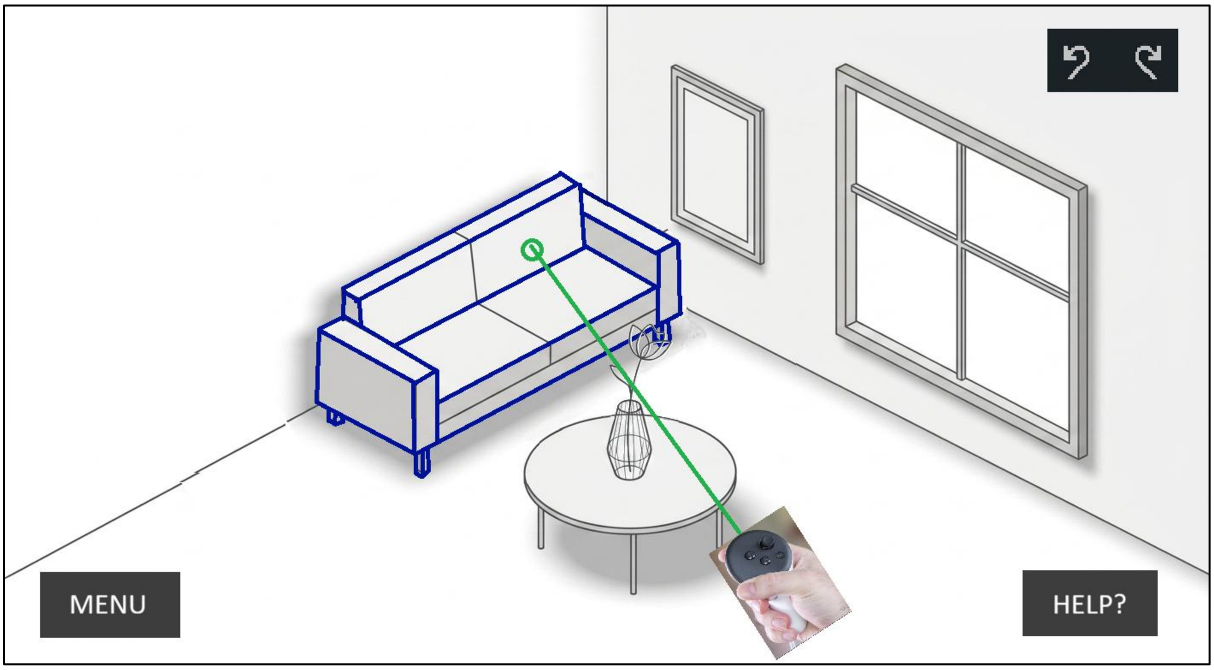
User goal:
Confirm the specific object they want to customise before moving to the editing and AI customisation flow.

2.2 Scene 2 — Move / Rotate Object

Interaction:
Click, hold & move / Rotate the selected object

Story:
After selecting an object, the user can directly adjust its position or orientation within the room. By clicking and dragging the selected object, the user can **move it to a new position**. A rotation control allows the user to **rotate the object** to the desired orientation.

The object's selection indicator remains visible while it is being manipulated so the user knows which object they are controlling.



User goal:

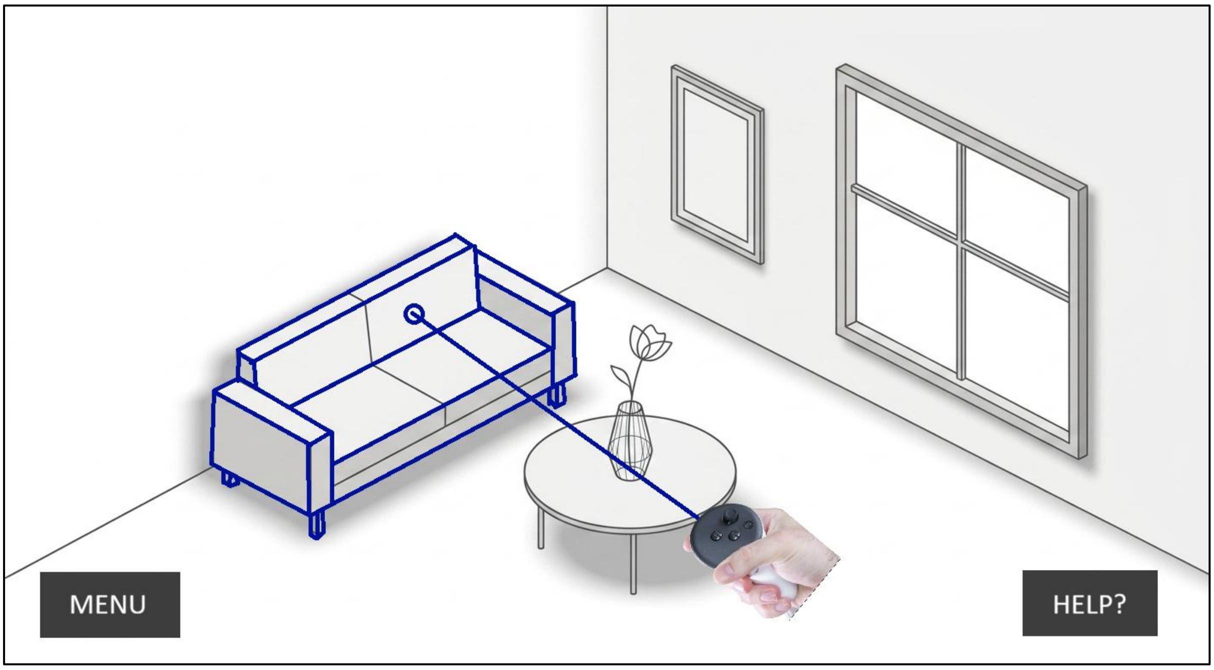
Position and orient the selected object within the interior to achieve the desired layout.

3. Movement / Rotation Constraint Warning

3.1 Scene 1: Select the Object

Interaction:
Click / Select the highlighted object

Story:
The user selects the highlighted object. The selected object changes from the **light highlight to a clear dark border**, confirming that the object has been successfully selected. The selection state makes it clear which object the user's next action will affect.

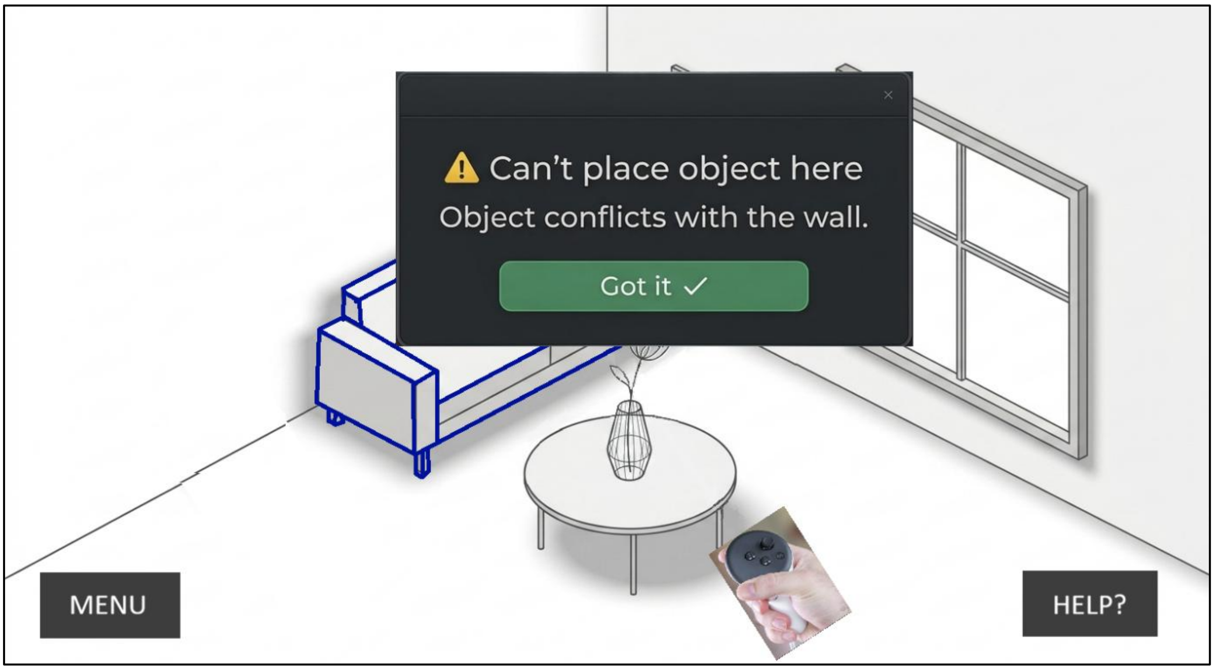


User goal:
Confirm the specific object they want to customise before moving to the editing and AI customisation flow.

3.2 Scene 2 — Collision Detected

Interaction:
Move / Rotate → Collision detected

Story:
When the user attempts to move or rotate the selected object toward a wall or another restricted area, the system detects the spatial constraint and provides **immediate visual feedback**. The object indicates that the proposed position or rotation is not valid, and a warning appears without blocking the user's view



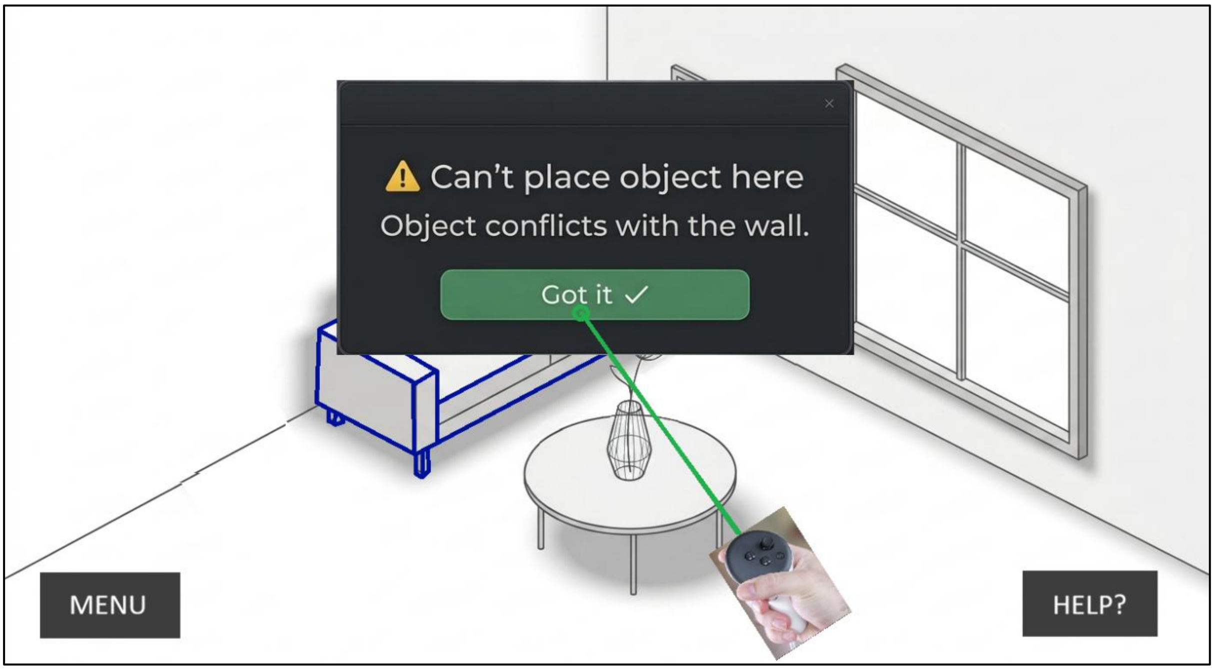
User goal:

Understand that the object cannot be moved or rotated into the current position because it conflicts with the room's spatial constraints

3.3 Scene 3: Acknowledge Warning

Interaction:
Click "OK" on the warning message

Story:
After reviewing the constraint warning, the user clicks **OK** to acknowledge the message. The warning panel closes, and the selected object remains in its last valid position. The user can then continue adjusting the object's position or rotation within the allowed space.



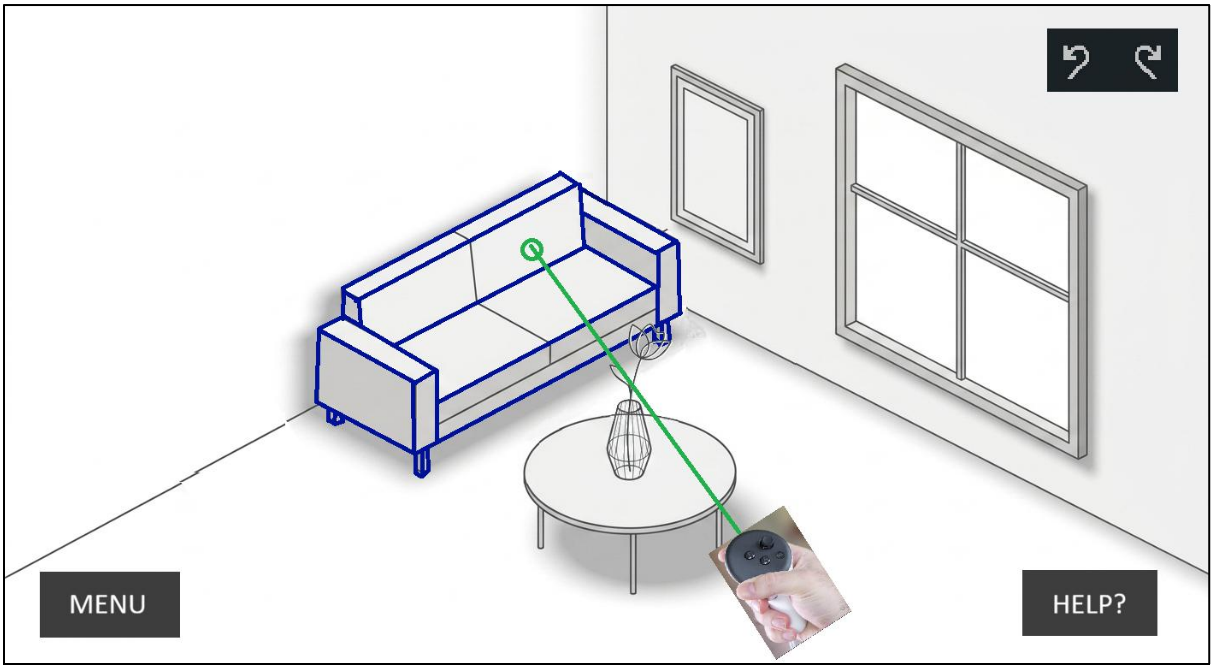
User goal:

Understand the constraint and continue editing the object without losing their current work.

3.4 Scene 4: Warning Acknowledged & Object Reverted

Interaction:
Click "OK" → Return to last valid position

Story:
After reviewing the constraint warning, the user clicks **OK** to acknowledge the message. The warning panel closes, and the selected object automatically returns to its **last valid position and orientation before the collision occurred**. The object remains selected, allowing the user to continue moving or rotating it within the available space.



User goal:

Understand the constraint and continue editing from the last valid placement without causing overlap or collision.

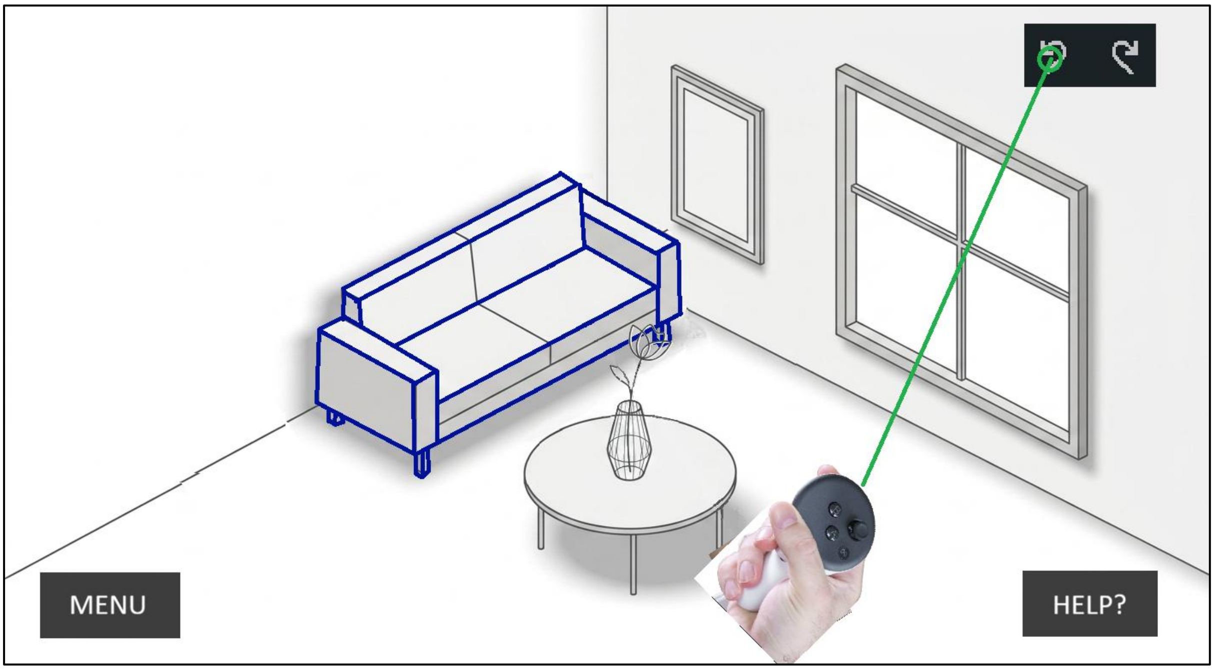
4. Undo Modification

4.1 Scene 1, 2, 3, 4 from section 3.

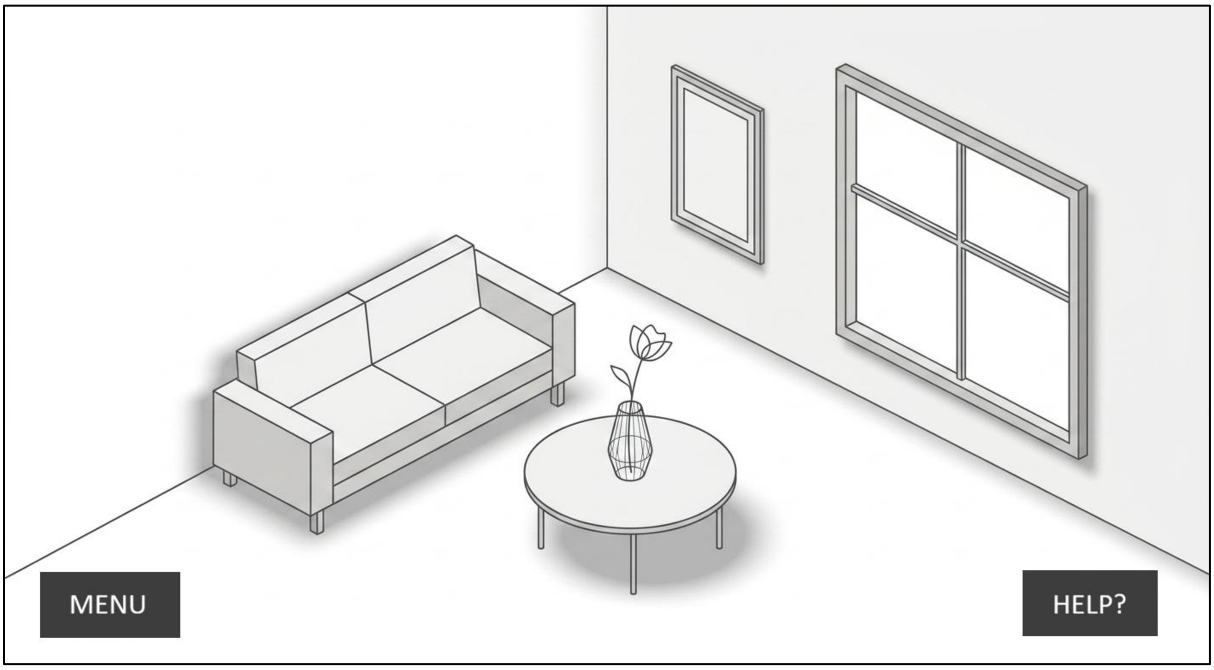
4.2 Scene 5: Undo Modification

Interaction:
Click Undo

Story:
The user is not satisfied with the generated result and selects **Undo**. The object returns to its previous state before the most recent modification was applied.



User goal: Revert an unwanted change quickly.



5. Menu Feature

5.1 Scene 1 — Initial State

When the application first starts, the **Menu** is in its collapsed state.

The bottom-left corner contains a single ☰ **Menu** button. The available menu options are hidden so that the main 3D scene remains clear and unobstructed.

At the top-right corner, an ╳ **Close** button is visible. This button allows the user to close the entire running application/section.

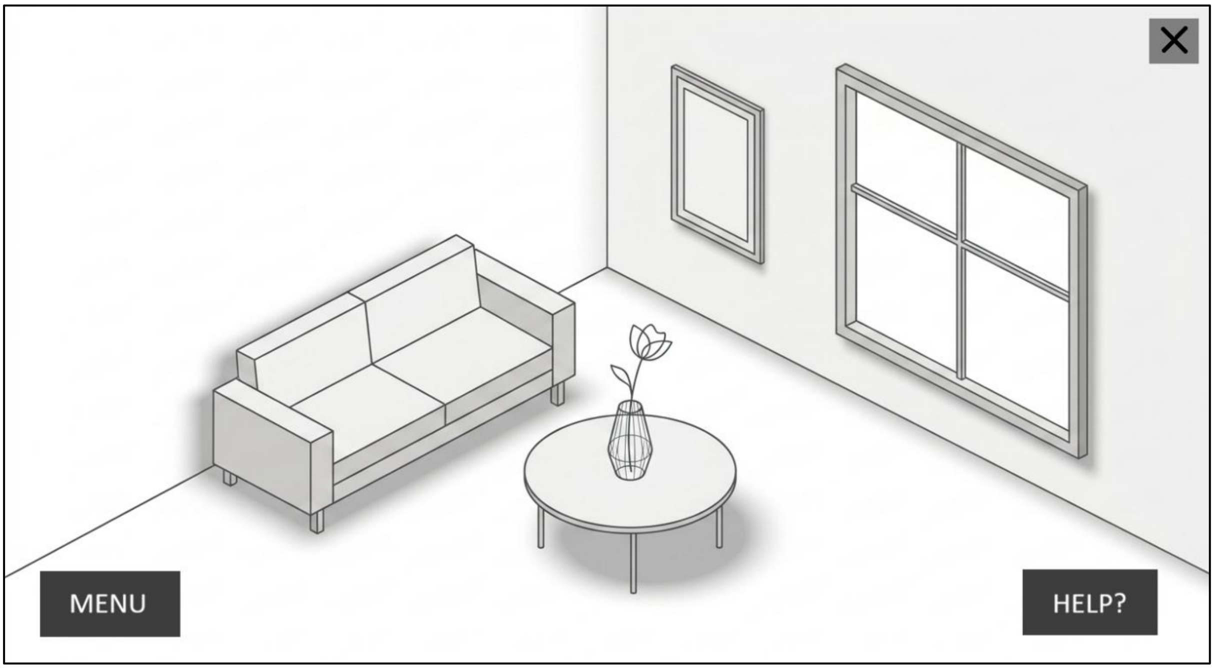
Purpose:

- Keep the interface clean when no menu actions are required.
- Provide quick access to additional functionality through the Menu button.
- Keep the close/exit functionality easily accessible.

Interaction:
Click "Menu" → Expand the menu options

Story:

Initially, the Menu remains collapsed in the bottom-left corner of the screen, displaying only the ☰ Menu button. The main 3D scene remains unobstructed, while the ╳ button in the top-right corner remains available to close the entire running section.



User goal:

Access additional features such as **New Item** and **Help** when needed, while keeping the main 3D scene clear and unobstructed when the menu is not in use.

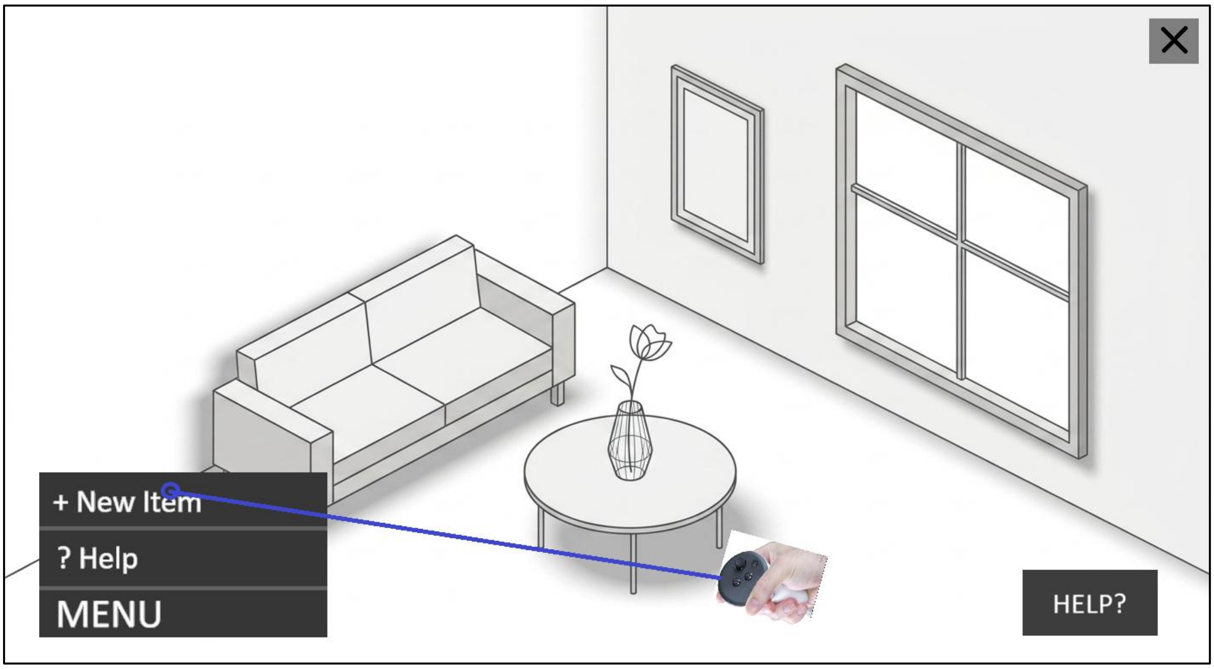
5.2 Scene 2 — Menu Expanded

Interaction:

Click "New Item" → Open the new item creation workflow

Story:

After clicking the **Menu** button, the available options are displayed in the bottom-left corner, including **New Item**, **Help**, and other future features. The user can select **New Item** to begin the process of adding a new item to the 3D scene. The menu remains available so the user can access other functions when needed.



User goal:

Access the **New Item** feature and begin adding a new item to the current 3D scene while keeping the available menu options easily accessible.

5.3 Scenes 4 – 7 from Section 1

6. Exit Application

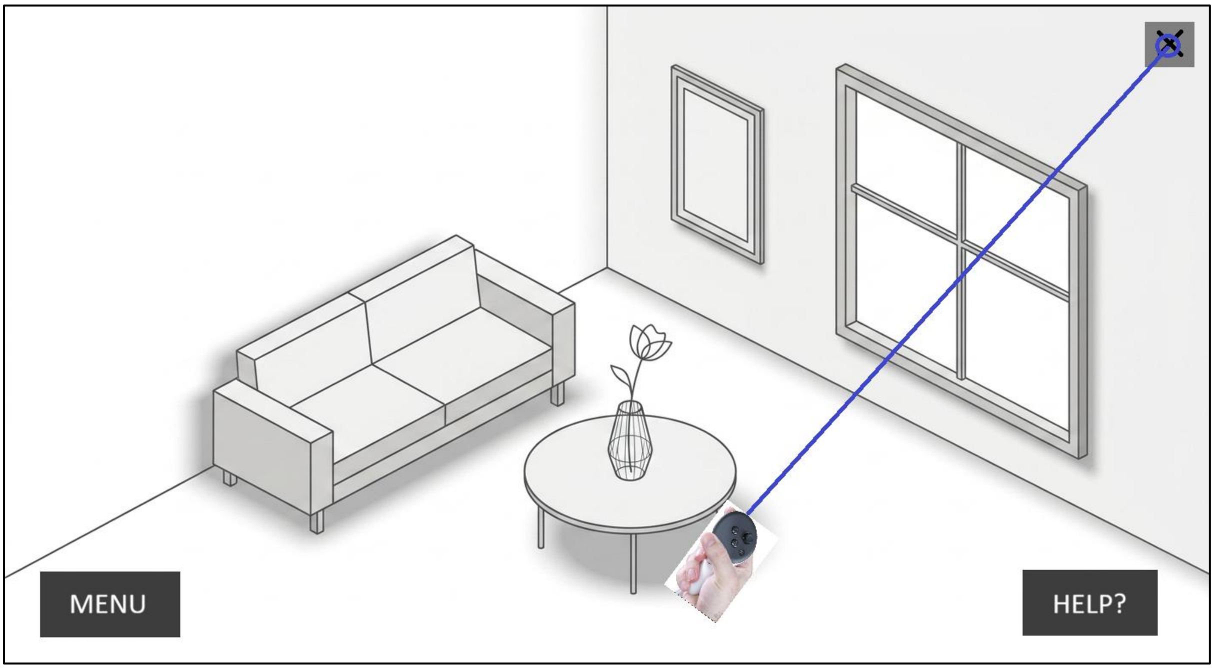
6.1 Exit Application

Interaction:

Click "✕" → Exit the application

Story:

The user clicks the ✕ button in the top-right corner to exit the entire running application. The system closes the current running section and returns the user to the appropriate environment outside the application.



User goal:

Exit the application safely and end the current session when they have finished using the 3D scene.

Revision History

Date	Version	Change
18/08/2026	1.0	Initial Version
07/09/2026	1.1	Following changes are made: 1. Added sections and sketch diagrams for Menu feature and exit the application. 2. Updated the image in section 1.6 with cancel option